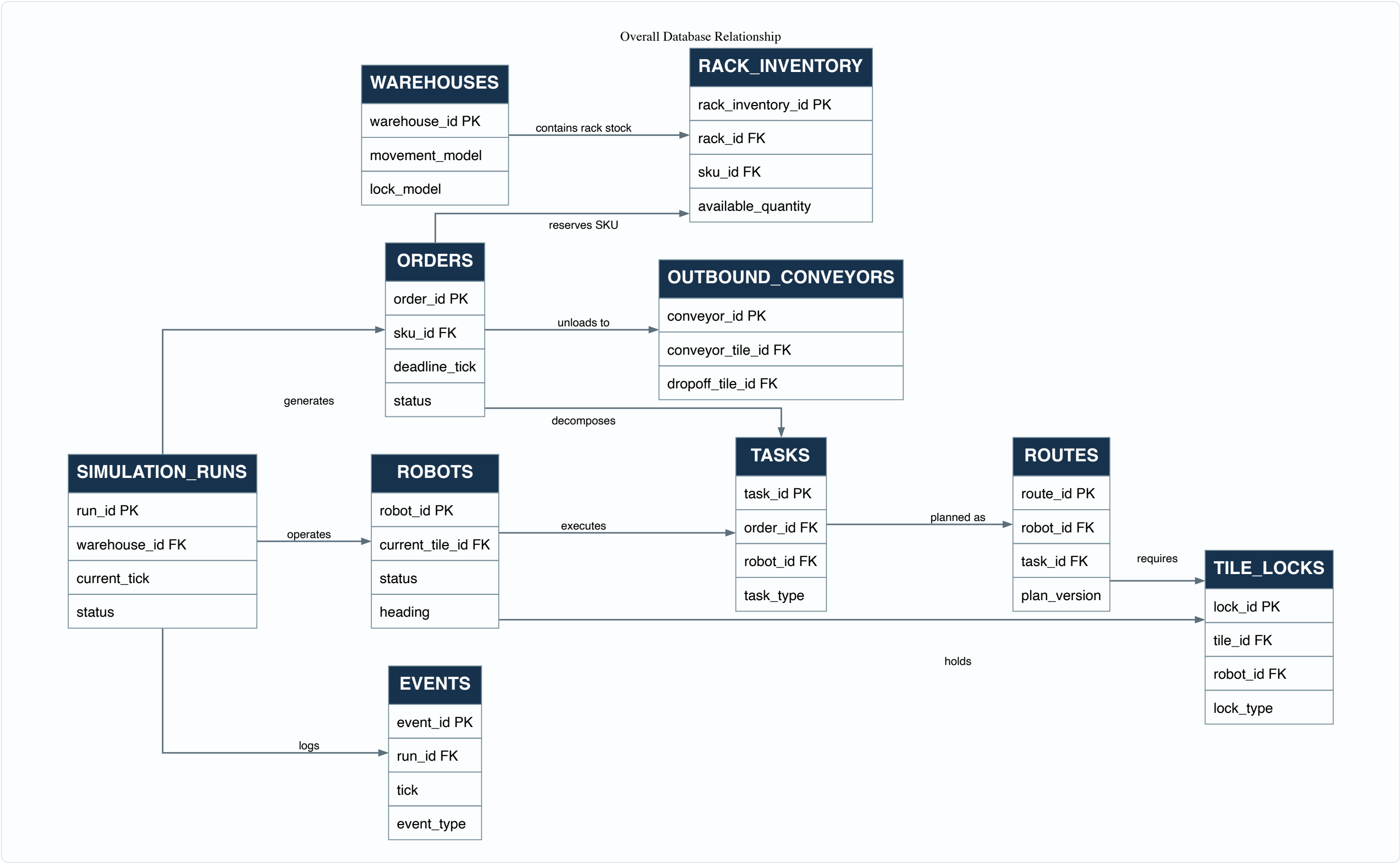


Warehouse Database Diagrams

Rendered database relationship diagrams for the outbound-only warehouse fulfillment simulator. The visual runtime should read map, rack, robot, tile lock, and outbound state from these database relationships.

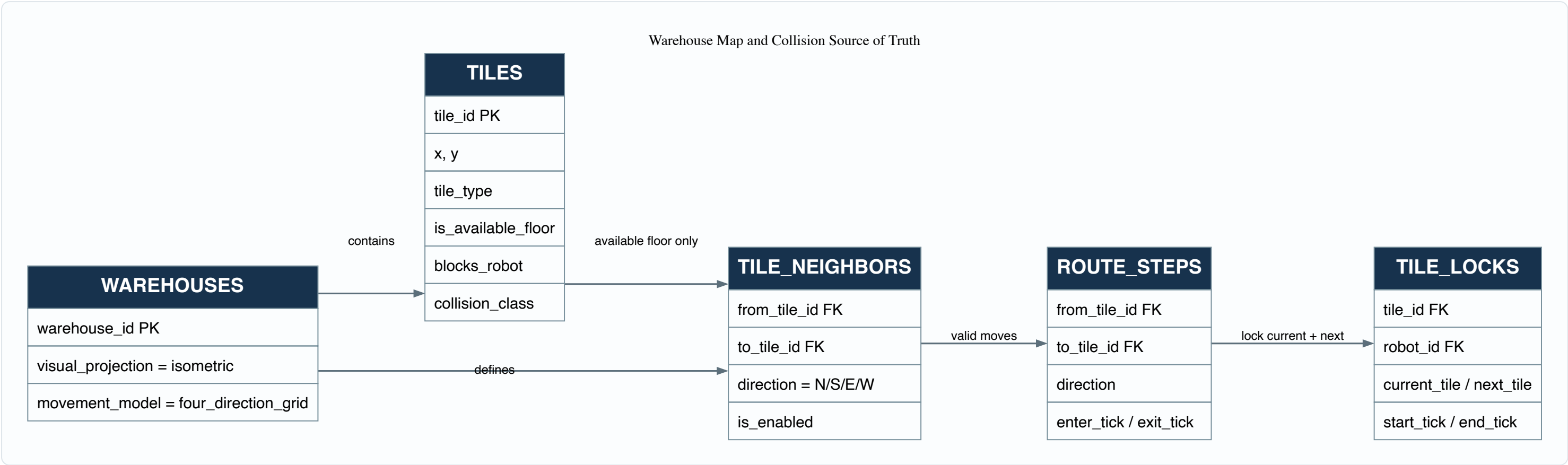
1. Overall Database Relationship

Warehouse runtime state flows from generated orders to inventory reservations, robot tasks, tile locks, conveyor unload, and completed orders.



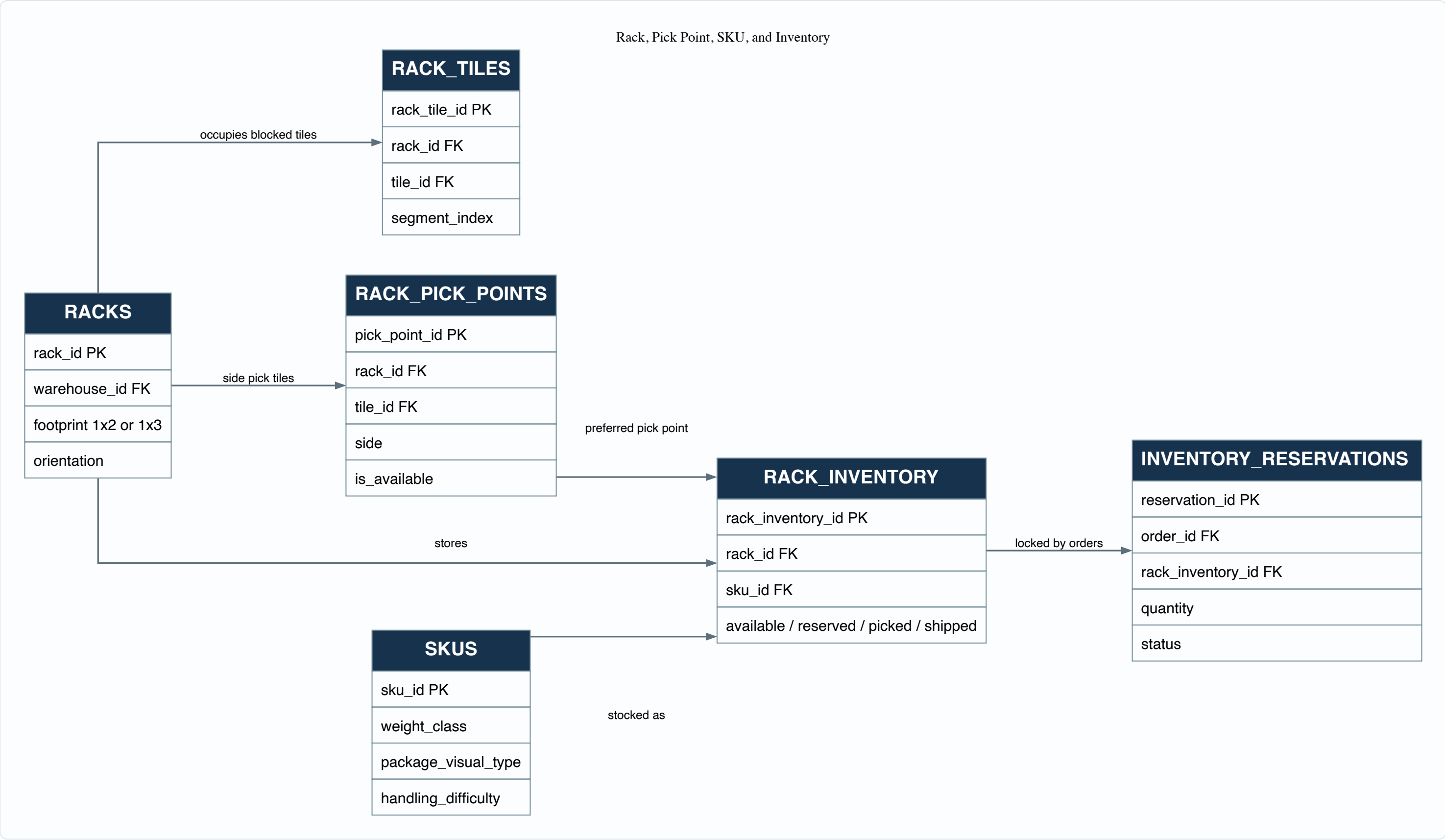
2. Warehouse Map, Available Floor, and Collision

Visual and pathfinding must use is_available_floor and tile_neighbors. Isometric 45-degree screen movement is only a projection of N/S/E/W grid movement.



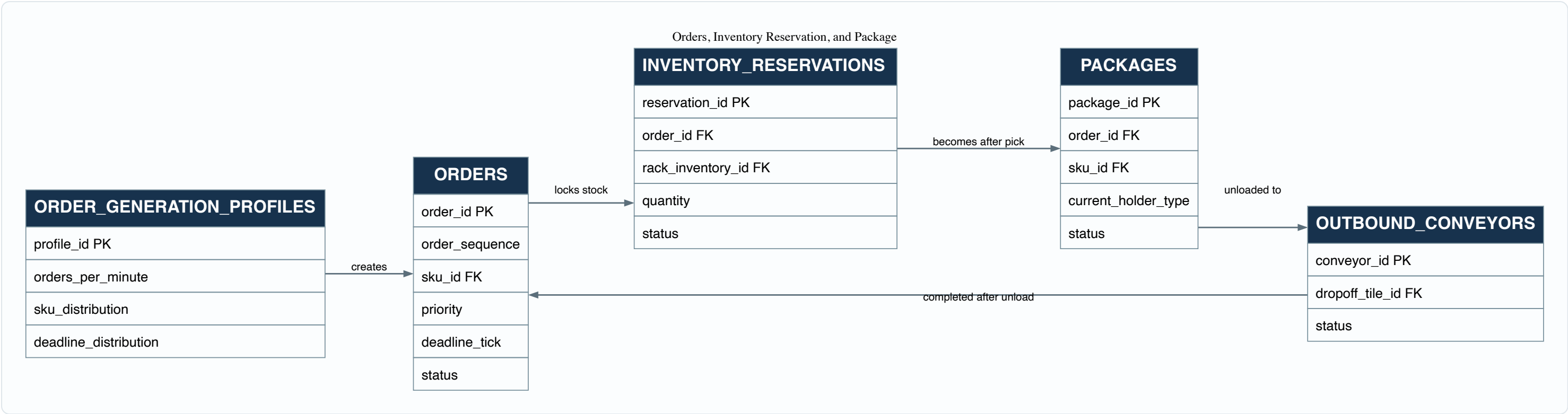
3. Rack, Pick Point, SKU, and Inventory

Racks occupy blocked tiles. Robots stand on side pick point tiles. SKU difficulty and weight class drive handling time and visual package type.



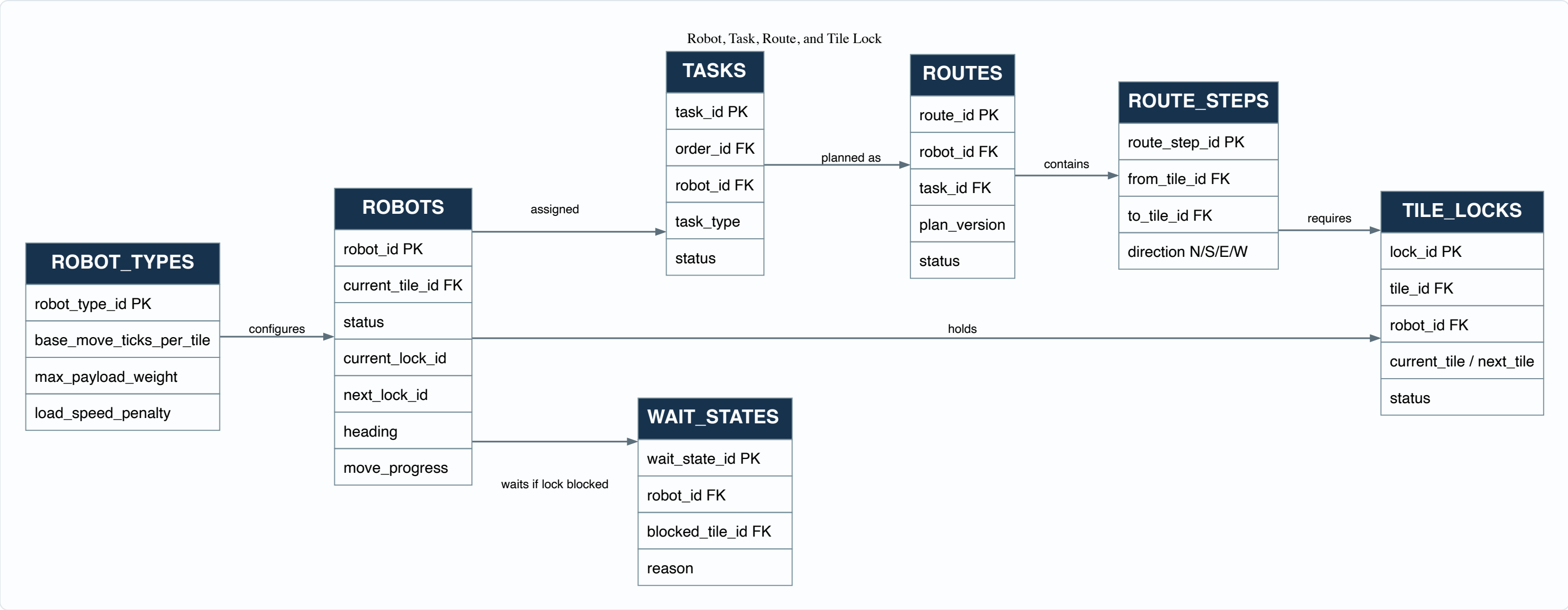
4. Orders, Inventory Reservation, and Package

Orders are auto-generated with long unique IDs. Inventory is outbound-only: available becomes reserved, picked, shipped.



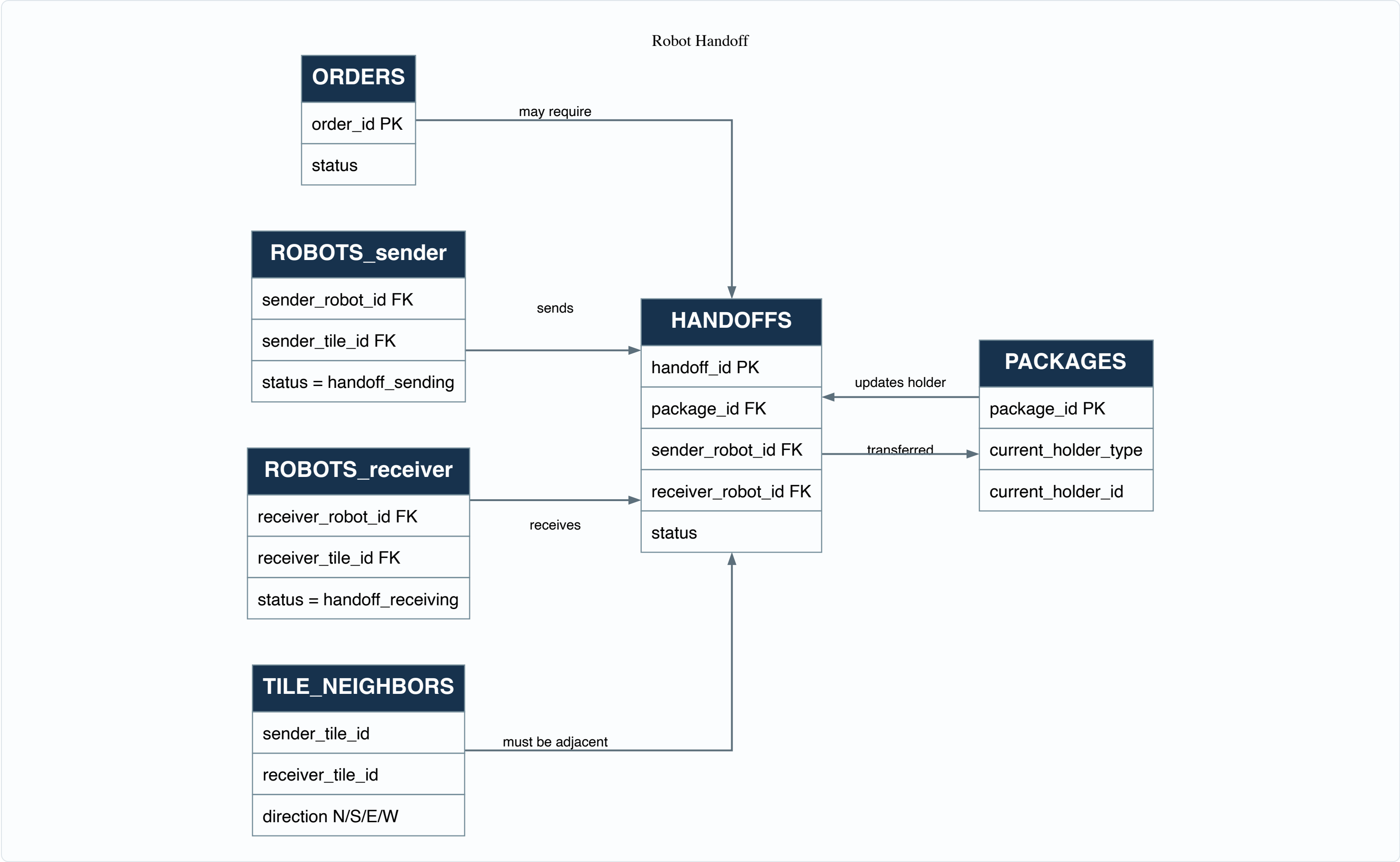
5. Robot, Task, Route, and Tile Lock

The collision model is simplified into tile locks: a robot owns its current tile and must reserve its next tile before moving.



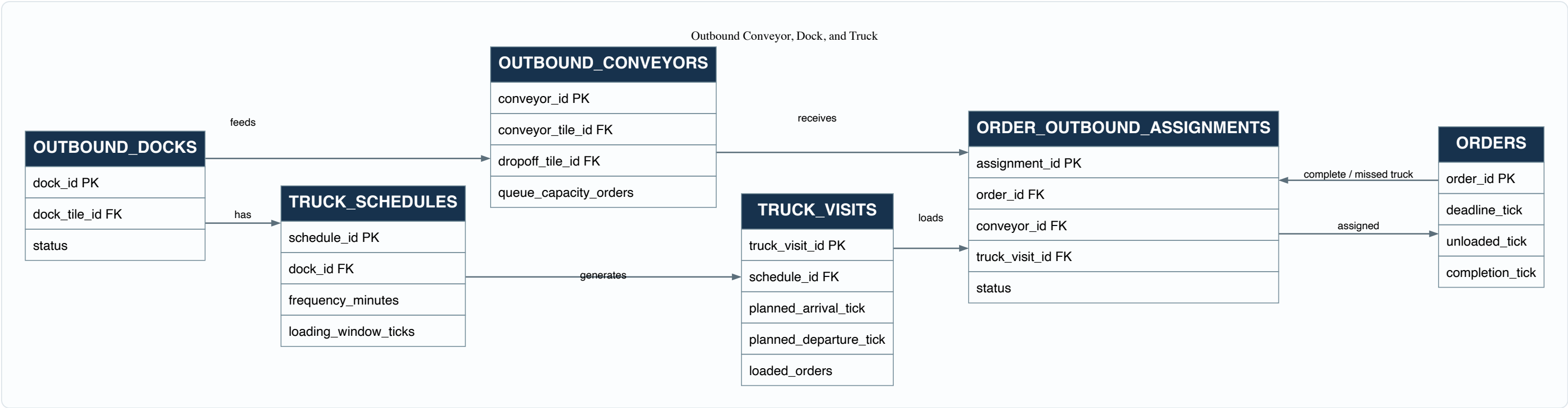
6. Robot Handoff

A handoff is valid only when sender and receiver robots occupy N/S/E/W adjacent available floor tiles.



7. Outbound Conveyor, Dock, and Truck

Robots do not enter conveyor tiles. They stand on dropoff_tile_id, unload to the conveyor, then the order can be counted as warehouse-completed.



8. Minimal First Implementation Chain

The first runnable database path only needs this chain to support order generation, rack depletion, robot movement, lock waiting, conveyor unload, and throughput calculation.

